MODULE 2: ENERGY DEVICES & SEMICONDUCTOR ELECTRODES

Official FAT Study Textbook & Step-by-Step Question Bank Solutions (CO: 2)

Aligned with Official VIT BACHY101 Syllabus, Department Question Bank, and Past FAT Papers

Scope of Module 2 (Official VIT Syllabus - All 9 Topics):

(1) Electrochemical and electrolytic cells, (2) Nernst equation, (3) Industrial electrolysis: Downs cell and water splitting, (4) Semiconductor electrodes, (5) Zone refining and single-crystal growth (Czochralski), (6) Lithium-ion battery chemistry and safety, (7) Fuel cells (PEMFC and SOFC), (8) Solar conversion: Silicon p-n and dye-sensitized solar cells (DSSC), and (9) Worked Question Bank Problems.

1. Core Syllabus Definitions & Theoretical Foundations (Line-by-Line)

1. Galvanic / Electrochemical Cell

A device that directly converts spontaneous chemical redox energy into usable electrical work ($\Delta G < 0, E_{\text{cell}} > 0$). Spontaneous oxidation occurs at the negative anode (LOAN rule: Left, Oxidation, Anode, Negative), while reduction occurs at the positive cathode.

2. Electrolytic Cell

An electrochemical device that consumes external electrical energy from an applied DC voltage source ($E_{\text{ext}} > E^{\circ}_{\text{cell}}$) to drive non-spontaneous chemical reactions ($\Delta G > 0$). The anode is connected to the positive terminal of the power supply and oxidation takes place there.

3. Salt Bridge

An inverted U-shaped glass tube filled with agar-agar gel saturated with inert electrolytes (such as KCl, KNO_3, or NH_4NO_3) where the cation and anion have virtually identical ionic mobilities ($u_+ \approx u_-$). It serves 3 essential functions: (1) Completes the internal electric circuit, (2) Maintains electroneutrality in both half-cells, and (3) Eliminates liquid junction potential ($E_j \approx 0$).

4. Nernst Equation

The fundamental thermodynamic relation expressing the electromotive force (EMF) of a cell as a function of standard reduction potential, temperature, and the reaction quotient Q:

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{RT}{nF} \ln Q = E^{\circ}_{\text{cell}} - \left(\frac{0.05916}{n}\right) \log_{10} Q \quad (@298.15\ \text{K})$$

5. Downs Cell Industrial Electrolysis

A continuous commercial electrochemical process for extracting pure metallic sodium by electrolyzing molten $NaCl$ at $\approx 600\,^{\circ}$ C (fluxed with $CaCl_2$ to lower melting point from $801\,^{\circ}$ C). A central cylindrical graphite anode produces dry Cl_2 gas and a surrounding circular steel cathode collects

molten sodium metal, separated by a cylindrical steel wire gauze diaphragm with operating cell voltage > 4.07 V.

6. Water Splitting Electrolysis
The stoichiometric electrochemical decomposition of water into hydrogen and oxygen gases in a $2:1$ molar ratio ($2H_2O \rightarrow 2H_2 + O_2$). Acidic mechanism uses platinum electrodes (Cathode : $2H^+ + 2e^- \rightarrow H_2$; Anode : $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$); alkaline mechanism uses nickel mesh electrodes with hydroxide ion migration.

7. Intrinsic Semiconductors
Extremely pure Group 14 elements (silicon, germanium) where electrical conductivity arises purely from thermal excitation of valence electrons across the bandgap E_g into the conduction band, creating equal concentrations of electrons and holes ($n_e = n_h = n_i$).

8. Extrinsic Semiconductors
Semiconductor crystals doped with controlled trace concentrations of impurity atoms:
- **n-type:** Doped with pentavalent Group 15 donors (P, As, Sb), producing mobile conduction electrons as majority carriers.
- **p-type:** Doped with trivalent Group 13 acceptors (B, Al, Ga), generating mobile positive electron holes in the valence band as majority carriers.

9. Three-Stage Silicon Purification
The rigorous industrial process used to manufacture semiconductor-grade silicon of $> 99.9999999\%$ (9N) purity:
- **Stage 1 (Siemens Process):** Carbothermic reduction of silica quartz followed by conversion into trichlorosilane ($SiHCl_3$), fractional distillation, and CVD reduction onto pure silicon filaments at $1100\,^\circ$ C.
- **Stage 2 (Zone Refining):** Repeated passage of a narrow RF induction molten zone along the silicon ingot sweeps impurities with segregation coefficient $k < 1$ into the terminal end, which is cropped off.
- **Stage 3 (Czochralski Single-Crystal Pulling):** Dipping an orienting seed crystal into molten silicon at $1425\,^\circ$ C and pulling slowly with rotation under an inert argon atmosphere to yield large dislocation-free single-crystal boules.

10. Lithium-Ion Secondary Battery (LIB)
A high-energy density rechargeable battery operating via the "rocking-chair" intercalation and de-intercalation of Li^+ ions:
- **Anode:** Graphitic carbon on copper foil (Li_xC_6).
- **Cathode:** Lithium cobalt oxide ($LiCoO_2$) or lithium iron phosphate ($LiFePO_4$) on aluminum foil.
- **Electrolyte:** 1 M $LiPF_6$ dissolved in organic alkyl carbonates (ethylene carbonate EC + dimethyl carbonate DMC).
- **Solid Electrolyte Interphase (SEI):** A passivating, protective nanometer film formed on the graphite anode surface during the initial charging cycle that conducts Li^+ ions while blocking electron transfer and electrolyte reduction.

11. Proton Exchange Membrane Fuel Cell (PEMFC)
A low-temperature ($60 - 80\,^\circ$ C) electrochemical power generator consuming pure hydrogen fuel and oxygen oxidant across a perfluorosulfonic acid ion-exchange membrane (Nafion):

- **Anode (Pt Catalyst):** $2H_2 \rightarrow 4H^+ + 4e^-$
- **Cathode (Pt Catalyst):** $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
- **Overall Reaction:** $2H_2 + O_2 \rightarrow 2H_2O + \text{Electricity} + \text{Heat}$

12. Solid Oxide Fuel Cell (SOFC)
A high-temperature ($700 - 1000\,^\circ$ C) all-solid-state ceramic fuel cell utilizing dense yttria-stabilized zirconia (YSZ) as an oxygen ion (O^{2-}) conducting solid electrolyte. Capable of directly utilizing syngas ($CO + H_2$) and reforming hydrocarbons without expensive noble metal catalysts.

13. Silicon p-n Junction Photovoltaic Cell
A semiconductor diode device that directly converts solar radiation into electricity. Photons with energy $h\nu \geq E_g$ absorb near the $p-n$ depletion zone, creating electron-hole pairs separated by the internal built-in electric field to produce open-circuit photovoltage (V_{oc}) and short-circuit photocurrent (I_{sc}).

14. Dye-Sensitized Solar Cell (DSSC / Grätzel Cell)
A biomimetic photoelectrochemical cell comprising: (1) Fluorine-doped tin oxide (FTO) conductive glass, (2) Nanoporous mesoporous TiO_2 photoanode, (3) Ruthenium polypyridyl dye sensitizer, (4) Liquid $\frac{I^-}{I_3^-}$ redox electrolyte mediator, and (5) Platinum counter-electrode.

2. In-Depth Chemical Theory & Step-by-Step Derivations

Thermodynamic Derivation of the Nernst Equation

From chemical thermodynamics, the change in Gibbs free energy for a general reversible redox process $aA + bB \Leftrightarrow cC + dD$ is given by the van 't Hoff reaction isotherm:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ is the reaction quotient. The maximum non-expansion electrical work obtained from a reversible galvanic cell is related to the Gibbs free energy change by:

$$\Delta G = -nFE_{\text{cell}} \quad \text{and} \quad \Delta G^\circ = -nFE^\circ_{\text{cell}}$$

where n is the number of moles of electrons transferred and $F = 96485$ C/mol is Faraday's constant. Substituting into the isotherm:

$$-nFE_{\text{cell}} = -nFE^\circ_{\text{cell}} + RT \ln Q$$

Dividing both sides by $-nF$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$$

Converting natural logarithm to base 10 ($\ln Q = 2.3026 \log_{10} Q$) and evaluating at standard temperature $T = 298.15$ K:

$$\frac{2.302585 \times 8.31446 \times 298.15}{96485} = \mathbf{0.05916}$$

Hence, the working Nernst equation is:

$$\boldsymbol{E_{\text{cell}} = E^{\circ}_{\text{cell}} - \left(\frac{0.05916}{n}\right) \log_{10} Q}$$

3. VIT FAT Exam Blueprint & Master Formula Sheet

Device / Law	Governing Equation / Formula	VIT Exam Key Insight
Nernst Cell EMF	$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \left(\frac{0.05916}{n}\right) \log_{10}\left(\frac{[\text{anode ions}]}{[\text{cathode ions}]}\right)$	$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$ (both reduction potentials).
Equilibrium Constant	$\log_{10} K_{\text{eq}} = \frac{n E^{\circ}_{\text{cell}}}{0.05916}$	At equilibrium, $E_{\text{cell}} = 0$ and $Q = K_{\text{eq}}$.
Faraday's First Law	$m = ZIt = \frac{MIt}{nF}$	1 Faraday = 96485 C; deposits 1 gram-equivalent.
Theoretical Cell Voltage	$\Delta G^{\circ} = -nFE^{\circ} \rightarrow E^{\circ} = -\frac{\Delta G^{\circ}}{nF}$	For water splitting ($\Delta G^{\circ} = +237.2$ kJ/mol): $E^{\circ} = 1.229$ V.
DSSC Energy Efficiency	$\eta = \frac{P_{\text{max}}}{P_{\text{in}}} = \frac{V_{\text{oc}} I_{\text{sc}}\ \text{FF}}{P_{\text{in}}}$	Fill Factor FF = $\frac{V_{\text{mp}} I_{\text{mp}}}{V_{\text{oc}} I_{\text{sc}}}$ (typically $0.70 - 0.85$).

4. Worked VIT Question Bank & Past FAT Numerical Problems

VIT FAT Problem 1 (10 Marks): Nernst Equation & Cell Potential Calculation

A galvanic electrochemical cell is represented by:

$$Zn(s) \mid Zn^{2+}(0.010\ \text{M}) \parallel Cu^{2+}(0.500\ \text{M}) \mid Cu(s)$$

Given standard reduction potentials $E^{\circ}\left(Z\frac{n^{2+}}{Z}n\right) = -0.763$ V and $E^{\circ}\left(C\frac{u^{2+}}{C}u\right) = +0.337$ V at 25 ° C.

(a) Write the half-cell reactions and the overall cell equation.
(b) Calculate standard cell potential E°_{cell}.
(c) Calculate the cell EMF E_{cell} under the given concentrations at 298 K.
(d) Determine the equilibrium constant K_{eq} of this Daniell cell.

Step-by-Step Solution:

- **Part (a): Cell Reactions ($n = 2$):**
  - Anode (Oxidation): $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-}$
  - Cathode (Reduction): $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$
  - Overall Cell Reaction: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
- **Part (b): Standard Cell Potential (E°_{cell}):**

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} = (+0.337) - (-0.763) = \mathbf{+1.100\ V}$$

- **Part (c): Cell EMF using Nernst Equation:**

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \left(\frac{0.05916}{2}\right) \log_{10}\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$$

$$\frac{[Zn^{2+}]}{[Cu^{2+}]} = \frac{0.010}{0.500} = 0.020 = 2.0 \times 10^{-2}$$

$$\log_{10}(0.020) = -1.6990$$

$$E_{\text{cell}} = 1.100 - (0.02958) \times (-1.6990) = 1.100 + 0.05025 = \mathbf{1.150\ V}$$

- **Part (d): Equilibrium Constant (K_{eq}):**

$$\log_{10} K_{\text{eq}} = \frac{nE^{\circ}_{\text{cell}}}{0.05916} = \frac{2 \times 1.100}{0.05916} = \frac{2.200}{0.05916} = 37.187$$

$$K_{\text{eq}} = 10^{37.187} = \mathbf{1.54 \times 10^{37}}$$

Boxed Final Answers:
(a) Standard Cell Potential $E^{\circ}_{\text{cell}} = \mathbf{+1.100\ V}$
(b) Cell EMF under operating conditions $E_{\text{cell}} = \mathbf{1.150\ V}$
(c) Thermodynamic Equilibrium Constant $K_{\text{eq}} = \mathbf{1.54 \times 10^{37}}$